

Theme

Use Streamlit's Settings menu (:
→ Settings) to switch between
the configured light and dark
themes.

MLOps

Feature Store

Feast Feature Store

Model Registry

MLflow Model Registry

Champion Model

Name: Pearls_AQI_XGBoost

Version: V15

Alias: champion

72-Hour Forecast Performance

Day 1

RMSE	MAE	R ²
6.72	3.13	0.949

Day 2

RMSE	MAE	R ²
6.72	3.15	0.949

Day 3

RMSE	MAE	R ²
6.71	3.15	0.949

Refresh dashboard

Pearls AQI Predictor • Peshawar

Pearls AQI Predictor

Peshawar Air Quality Monitoring & 3-Day Forecast

Current Air Quality

Current AQI

178

Unhealthy

Last Updated

07 Sep 2026, 05:00 PM PKT

Location: Peshawar

Current Pollutants

PM2.5

37.0

µg/m³

PM10

52.2

µg/m³

O₃

167.0

µg/m³

NO₂

12.4

µg/m³

SO₂

6.6

µg/m³

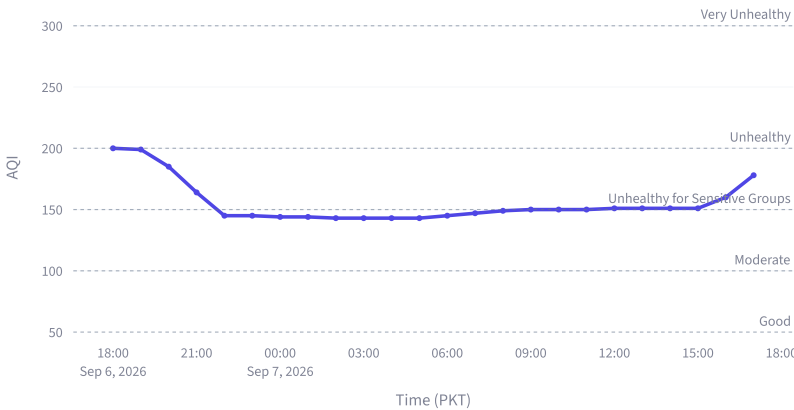
CO

534.0

µg/m³

Current 24-Hour AQI Trend

AQI — Last 24 Hours



Next 3-Day AQI Forecast for Peshawar

Day 1

2026-09-08

Predicted AQI

144.2

Unhealthy for Sensitive Groups

Champion RMSE vs Persistence ⓘ
+67.25%

Unhealthy for Sensitive Groups — Health Alert

Sensitive groups may experience health effects.

Recommendation: Sensitive individuals should consider reducing prolonged or heavy outdoor exertion.

Day 2

2026-09-09

Predicted AQI

159.0

Unhealthy

Champion RMSE vs Persistence ⓘ
+75.17%

Unhealthy — Health Alert

Everyone may begin to experience health effects.

Recommendation: Consider reducing prolonged or heavy exertion.

Day 3

2026-09-10

Predicted AQI

161.0

Unhealthy

Champion RMSE vs Persistence ⓘ
+77.13%

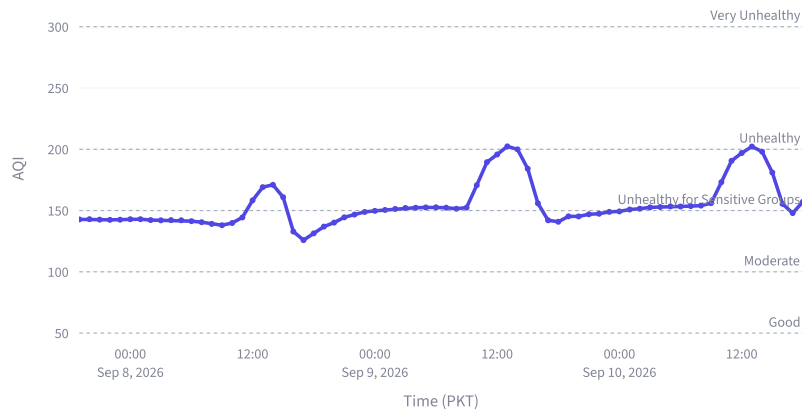
Unhealthy — Health Alert

Everyone may begin to experience health effects.

Recommendation: Consider reducing prolonged or heavy exertion.

Predicted AQI — Next 72 Hours

Predicted AQI — Next 72 Hours



The forecast shown here represents the next 72 hours (3 days), excluding the current day.

Why Is the AQI Expected to Be This Way?

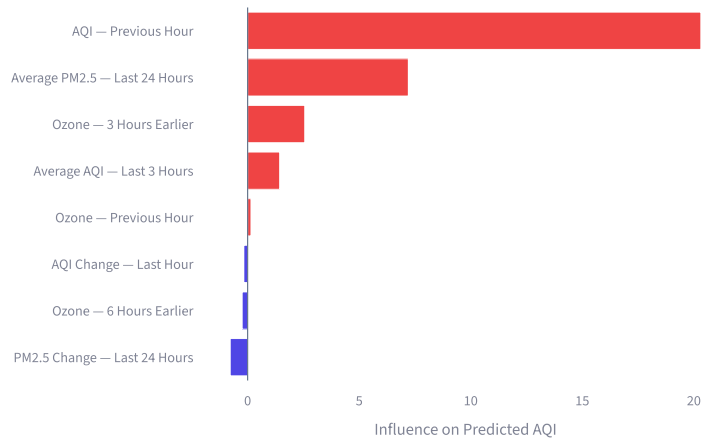
The charts below show the main factors that influenced each day's predicted AQI.

- Orange bars indicate factors that generally pushed the predicted AQI higher.
- Blue bars indicate factors that generally pushed the predicted AQI lower.

The longer the bar, the stronger the influence on the prediction.

Day 1 — 2026-09-08

Main Factors Influencing the 2026-09-08 Forecast



What influenced this forecast?

- Factors pushing AQI higher
 - AQI — Previous Hour (+20.31)
 - Average PM2.5 — Last 24 Hours (+7.20)
 - Ozone — 3 Hours Earlier (+2.56)
- Factors pushing AQI lower
 - AQI Change — Last Hour (-0.17)
 - PM2.5 Change — Last 24 Hours (-0.78)
 - Ozone — 6 Hours Earlier (-0.24)

What this means: Average PM2.5 — Last 24 Hours was one of the strongest factors pushing the predicted AQI higher on 2026-09-08.

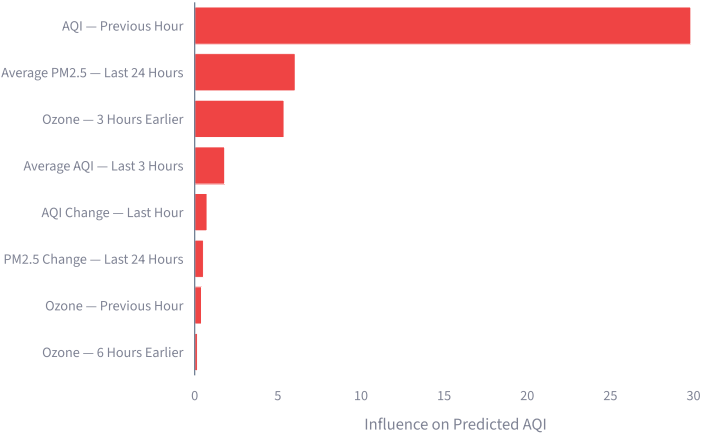
View SHAP feature details

Technical Feature	Dashboard Feature	Mean SHAP	Mean Absolute SHAP
aqi_lag_1	AQI — Previous Hour	20.3128	20.3128
pm2_5_24h_mean	Average PM2.5 — Last 24 Hours	7.2033	7.2033
aqi_change_1h	AQI Change — Last Hour	-0.1707	3.4714
ozone_lag_3	Ozone — 3 Hours Earlier	2.5592	3.4149
aqi_3h_mean	Average AQI — Last 3 Hours	1.4353	1.4353
pm2_5_change_24h	PM2.5 Change — Last 24 Hours	-0.7793	0.9215
ozone_lag_6	Ozone — 6 Hours Earlier	-0.2443	0.5935
ozone_lag_1	Ozone — Previous Hour	0.1501	0.5546
ozone	Ozone	0.0145	0.2946
pm10_change_24h	PM10 Change — Last 24 Hours	-0.2314	0.2715

These explanations summarize the factors considered by the prediction model. They do not mean that one factor alone determines AQI.

▼ Day 2 — 2026-09-09

Main Factors Influencing the 2026-09-09 Forecast



What influenced this forecast?

● Factors pushing AQI higher

- AQI — Previous Hour (+29.84)
- Ozone — 3 Hours Earlier (+5.38)
- Average PM2.5 — Last 24 Hours (+6.05)

● Factors pushing AQI lower

- Nitrogen Dioxide — Previous Day (-0.08)
- Carbon Monoxide (-0.11)
- Ozone — Previous Day (-0.06)

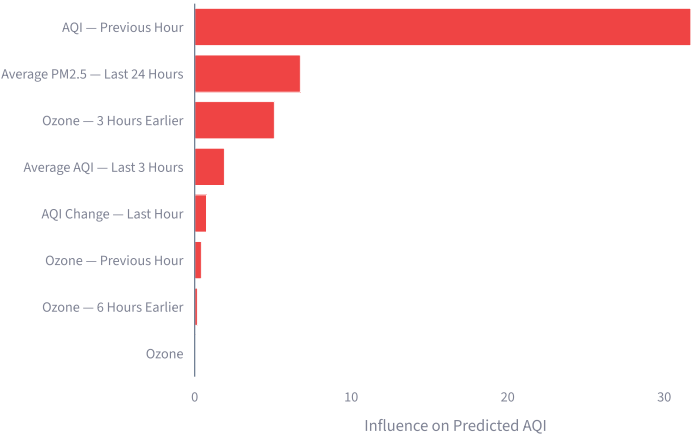
What this means: Ozone — 3 Hours Earlier was one of the strongest factors pushing the predicted AQI higher on 2026-09-09.

▼ View SHAP feature details

Technical Feature	Dashboard Feature	Mean SHAP	Mean Absolute SHAP
aqi_lag_1	AQI — Previous Hour	29.8382	29.8382
ozone_lag_3	Ozone — 3 Hours Earlier	5.3753	6.209
pm2_5_24h_mean	Average PM2.5 — Last 24 Hours	6.0482	6.0482
aqi_change_1h	AQI Change — Last Hour	0.7455	4.4135
aqi_3h_mean	Average AQI — Last 3 Hours	1.8039	1.8039
ozone_lag_1	Ozone — Previous Hour	0.4105	0.7954
ozone_lag_6	Ozone — 6 Hours Earlier	0.1667	0.6685
pm2_5_change_24h	PM2.5 Change — Last 24 Hours	0.5273	0.5659
ozone	Ozone	0.0554	0.3078
aqi_change_3h	AQI Change — Last 3 Hours	0.0263	0.2184

These explanations summarize the factors considered by the prediction model. They do not mean that one factor alone determines AQI.

Main Factors Influencing the 2026-09-10 Forecast



What influenced this forecast?

- Factors pushing AQI higher

 - AQI — Previous Hour (+31.70)
 - Average PM2.5 — Last 24 Hours (+6.75)
 - Ozone — 3 Hours Earlier (+5.09)
- Factors pushing AQI lower

 - AQI Change — Last 3 Hours (-0.00)
 - AQI Change — Last 6 Hours (-0.00)
 - Nitrogen Dioxide — Previous Day (-0.08)

What this means: Average PM2.5 — Last 24 Hours was one of the strongest factors pushing the predicted AQI higher on 2026-09-10.

View SHAP feature details

Technical Feature	Dashboard Feature	Mean SHAP	Mean Absolute SHAP
aqi_lag_1	AQI — Previous Hour	31.7014	31.7014
pm2_5_24h_mean	Average PM2.5 — Last 24 Hours	6.7538	6.7538
ozone_lag_3	Ozone — 3 Hours Earlier	5.0931	5.9764
aqi_change_1h	AQI Change — Last Hour	0.7558	3.5132
aqi_3h_mean	Average AQI — Last 3 Hours	1.9003	1.9003
ozone_lag_1	Ozone — Previous Hour	0.4294	0.789
ozone_lag_6	Ozone — 6 Hours Earlier	0.1751	0.6317
ozone	Ozone	0.0594	0.2578
aqi_change_3h	AQI Change — Last 3 Hours	-0.0045	0.2096
pm2_5_change_24h	PM2.5 Change — Last 24 Hours	0.1054	0.2094

These explanations summarize the factors considered by the prediction model. They do not mean that one factor alone determines AQI.